

PRAKTIKUM SISTEM OPERASI

MODUL 13

Perintah Dasar Linux



Oleh:

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2311104058

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

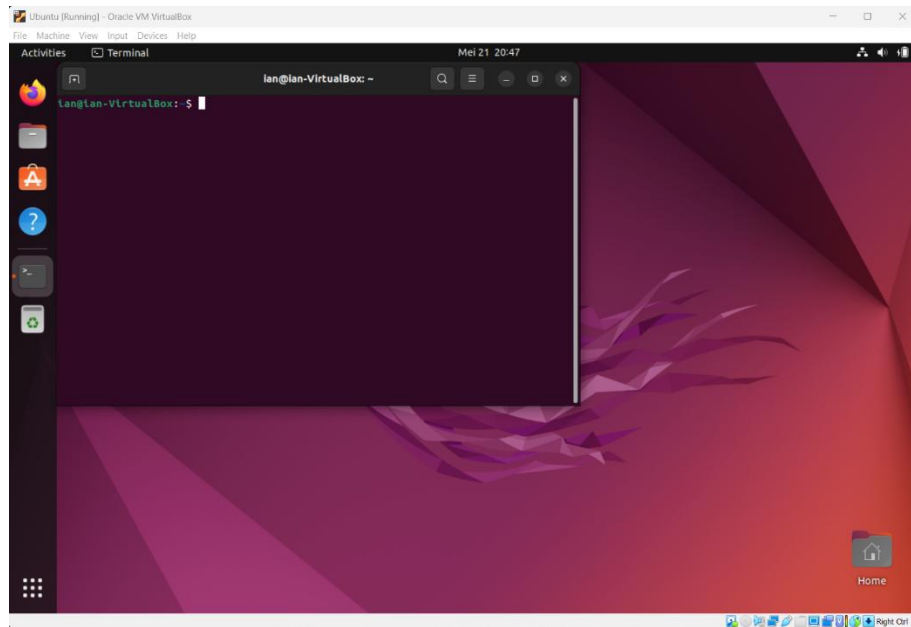
2026

JURNAL

I. Perintah-perintah Dasar Linux

1. Terminal

a. Screenshot

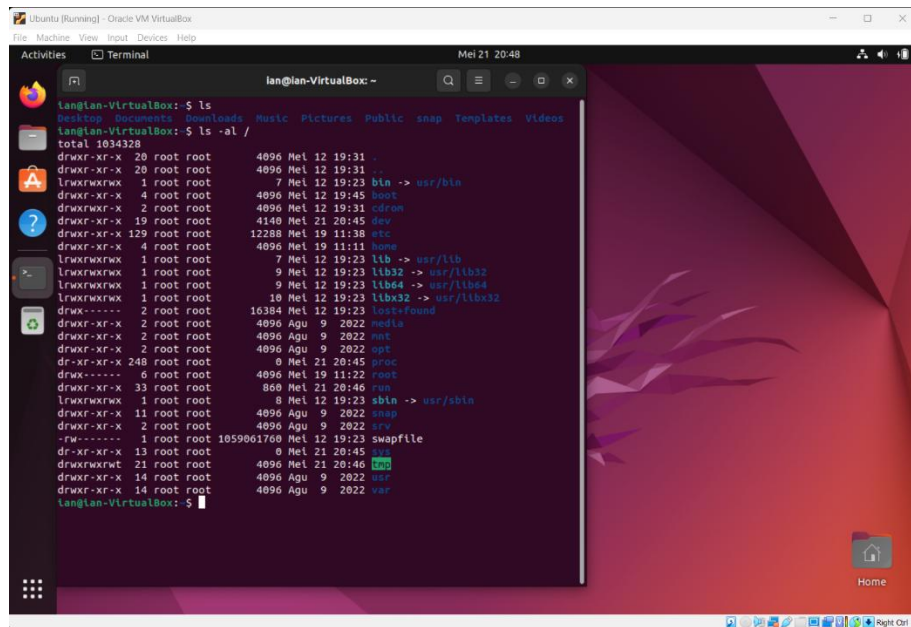


b. Arti dari command prompt

ian menunjukkan username yang sedang aktif atau login saat ini, ian-VirtualBox menunjukkan host atau nama komputer tempat user tersebut berada, ~ menunjukkan working directory atau direktori tempat berada saat ini yang dalam hal ini adalah direktori home, dan \$ menandakan bahwa user yang mengeksekusi terminal adalah user biasa, bukan root

2. Perintah Pertama

a. Screenshot



```
lan@lan-VirtualBox: ~  
$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
lan@lan-VirtualBox: ~$ ls -al /  
total 1034328  
drwxr-xr-x 20 root root      4096 Mei 12 19:31 .  
drwxr-xr-x 20 root root      4096 Mei 12 19:31 ..  
lrwxrwxrwx 1 root root         7 Mei 12 19:23 bin -> usr/bin  
drwxr-xr-x 4 root root      4096 Mei 12 19:45 boot  
drwxr-xr-x 2 root root      4096 Mei 12 19:31 cdrom  
drwxr-xr-x 19 root root     4140 Mei 21 20:45 dev  
drwxr-xr-x 129 root root    12288 Mei 19 11:38 etc  
drwxr-xr-x 4 root root      4096 Mei 19 11:11 home  
lrwxrwxrwx 1 root root         7 Mei 12 19:23 lib -> usr/lib  
lrwxrwxrwx 1 root root        10 Mei 12 19:23 lib32 -> usr/lib32  
lrwxrwxrwx 1 root root        10 Mei 12 19:23 lib64 -> usr/lib64  
lrwxrwxrwx 1 root root        10 Mei 12 19:23 libx32 -> usr/libx32  
drwx----- 2 root root    16384 Mei 12 19:23 lost+found  
drwxr-xr-x 2 root root      4096 Agu  9 2022 media  
drwxr-xr-x 2 root root      4096 Agu  9 2022 net  
drwxr-xr-x 2 root root      4096 Agu  9 2022 opt  
dr-xr-xr-x 248 root root       0 Mei 21 20:45 proc  
drwx----- 6 root root      4096 Mei 19 11:22 root  
dr-xr-xr-x 33 root root      860 Mei 21 20:46 run  
lrwxrwxrwx 1 root root         8 Mei 12 19:23 sbin -> usr/sbin  
drwxr-xr-x 11 root root      4096 Agu  9 2022 snap  
drwxr-xr-x 2 root root      4096 Agu  9 2022 srv  
----- 1 root root 1059061760 Mei 12 19:23 swapfile  
dr-xr-xr-x 13 root root       0 Mei 21 20:45 sys  
drwxrwxrwt 21 root root      4096 Mei 21 20:46 tmp  
drwxr-xr-x 14 root root      4096 Agu  9 2022 usr  
drwxr-xr-x 14 root root      4096 Agu  9 2022 var  
lan@lan-VirtualBox: ~$
```

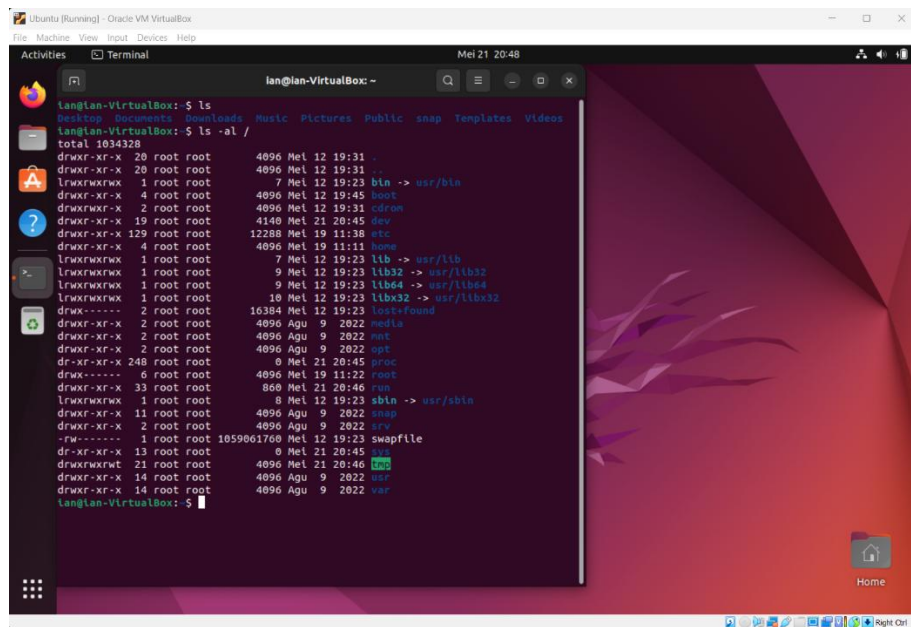
b. Option dan Parameter

Tidak ada option dan parameter

c. Fungsi

Menampilkan daftar (list) file dan direktori pada direktori aktif saat ini (secara bawaan tanpa file tersembunyi).

d. Screenshot



```
lan@lan-VirtualBox: ~  
$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
lan@lan-VirtualBox: ~$ ls -al /  
total 1034328  
drwxr-xr-x 20 root root      4096 Mei 12 19:31 .  
drwxr-xr-x 20 root root      4096 Mei 12 19:31 ..  
lrwxrwxrwx 1 root root         7 Mei 12 19:23 bin -> usr/bin  
drwxr-xr-x 4 root root      4096 Mei 12 19:45 boot  
drwxr-xr-x 2 root root      4096 Mei 12 19:31 cdrom  
drwxr-xr-x 19 root root     4140 Mei 21 20:45 dev  
drwxr-xr-x 129 root root    12288 Mei 19 11:38 etc  
drwxr-xr-x 4 root root      4096 Mei 19 11:11 home  
lrwxrwxrwx 1 root root         7 Mei 12 19:23 lib -> usr/lib  
lrwxrwxrwx 1 root root        10 Mei 12 19:23 lib32 -> usr/lib32  
lrwxrwxrwx 1 root root        10 Mei 12 19:23 lib64 -> usr/lib64  
lrwxrwxrwx 1 root root        10 Mei 12 19:23 libx32 -> usr/libx32  
drwx----- 2 root root    16384 Mei 12 19:23 lost+found  
drwxr-xr-x 2 root root      4096 Agu  9 2022 media  
drwxr-xr-x 2 root root      4096 Agu  9 2022 net  
drwxr-xr-x 2 root root      4096 Agu  9 2022 opt  
dr-xr-xr-x 248 root root       0 Mei 21 20:45 proc  
drwx----- 6 root root      4096 Mei 19 11:22 root  
dr-xr-xr-x 33 root root      860 Mei 21 20:46 run  
lrwxrwxrwx 1 root root         8 Mei 12 19:23 sbin -> usr/sbin  
drwxr-xr-x 11 root root      4096 Agu  9 2022 snap  
drwxr-xr-x 2 root root      4096 Agu  9 2022 srv  
----- 1 root root 1059061760 Mei 12 19:23 swapfile  
dr-xr-xr-x 13 root root       0 Mei 21 20:45 sys  
drwxrwxrwt 21 root root      4096 Mei 21 20:46 tmp  
drwxr-xr-x 14 root root      4096 Agu  9 2022 usr  
drwxr-xr-x 14 root root      4096 Agu  9 2022 var  
lan@lan-VirtualBox: ~$
```

e. Option dan Parameter

Option -al dan Parameter / (direktori root)

f. Fungsi

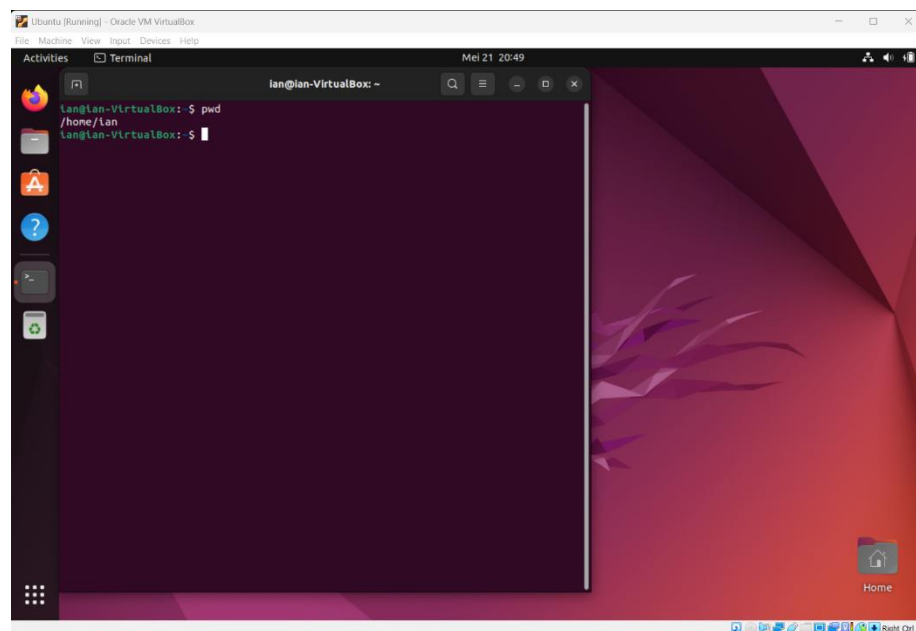
Menampilkan seluruh daftar isi yang ada di dalam direktori root (/) dengan format detail (list panjang), termasuk menampilkan file-file yang tersembunyi (hidden files).

g. Perbedaan perintah a dan e

Karena pada perintah ls biasa, terminal hanya menampilkan isi direktori yang sedang aktif saja tanpa format detail. Sedangkan pada perintah ls -al /, secara spesifik meminta terminal menampilkan isi dari direktori utama atau root (/) menggunakan parameter /, ditambah option -al yang memaksa terminal memunculkan detail hak akses, pemilik file, serta file yang disembunyikan.

3. Pohon File

a. Screenshot



b. Option dan Parameter

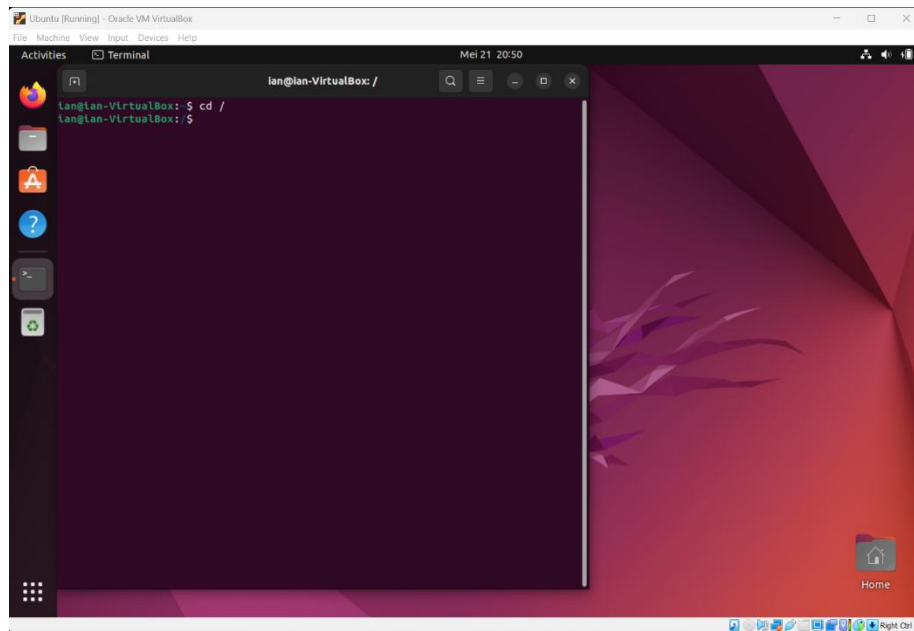
Tidak ada option dan Parameter

c. Fungsi

Singkatan dari Print Working Directory, berfungsi untuk menampilkan absolute path atau jalur direktori aktif tempat kamu berada saat ini (misalnya /home/ian).

4. Perpindahan

a. Screenshot



b. Option dan Parameter

Option tidak ada dan parameter /

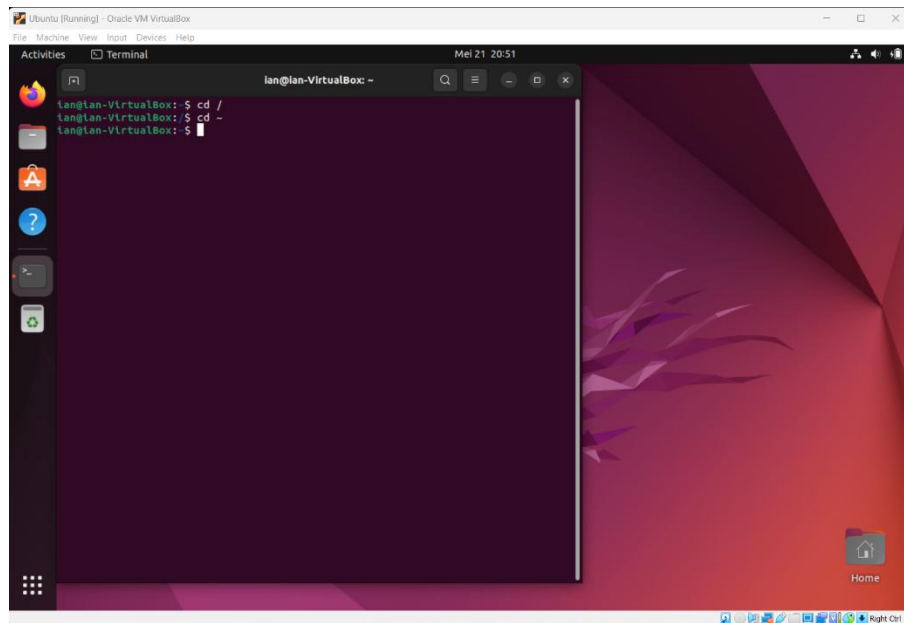
c. Yang dilakukan perintah

Memindahkan direktori aktif ke dalam direktori root (direktori paling awal/utama dalam struktur file Linux).

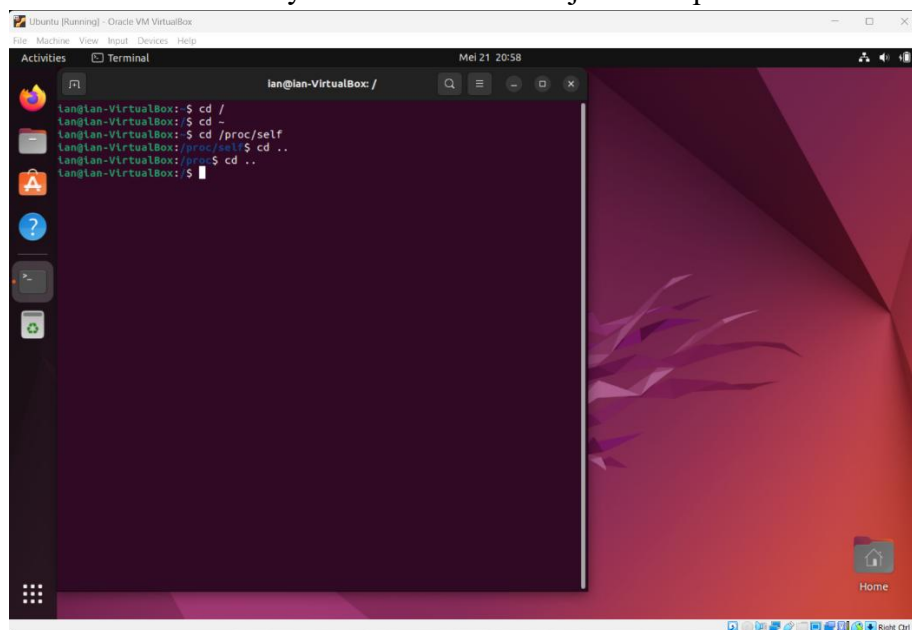
5. Direktori Khusus

a. Screenshot dan penjelasan hasil

Perintah `cd /` memindahkanmu ke direktori root. Setelah itu, eksekusi `cd ~` akan langsung menarik kembali ke direktori home milik user(yaitu `/home/ian`)

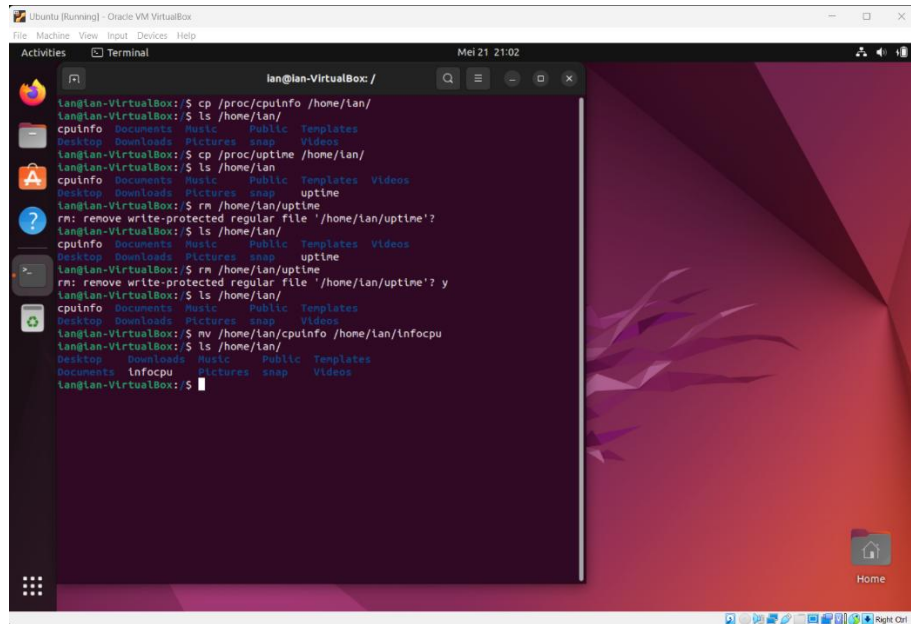


- b. Screenshot dan hasil
`cd ..` dieksekusi sebanyak 2 kali atau `cd ../../` jika dari `/proc/self`



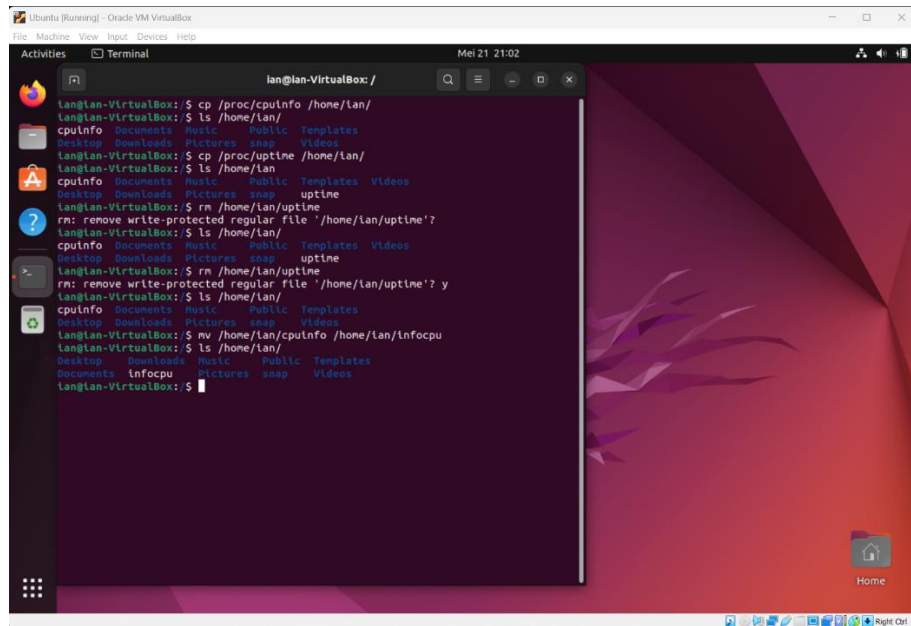
6. Copy, rename dan delete file

- Copy file /proc/cpuinfo
cp /proc/cpuinfo /home/ian
- Screenshot



```
ian@ian-VirtualBox: /
langlan-VirtualBox: $ cp /proc/cpuinfo /home/ian/
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates
Desktop  Downloads  Pictures  snap  Videos
langlan-VirtualBox: $ cp /proc/uptime /home/ian/
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates  Videos
Desktop  Downloads  Pictures  snap  uptime
langlan-VirtualBox: $ rm /home/ian/uptime
rm: remove write-protected regular file '/home/ian/uptime'?
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates  Videos
Desktop  Downloads  Pictures  snap  uptime
langlan-VirtualBox: $ rm /home/ian/uptime
rm: remove write-protected regular file '/home/ian/uptime'? y
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates
Desktop  Downloads  Pictures  snap  Videos
langlan-VirtualBox: $ mv /home/ian/cpuinfo /home/ian/infocpu
langlan-VirtualBox: $ ls /home/ian/
Desktop  Downloads  Music  Public  Templates
Documents  infocpu  Pictures  snap  Videos
langlan-VirtualBox: $
```

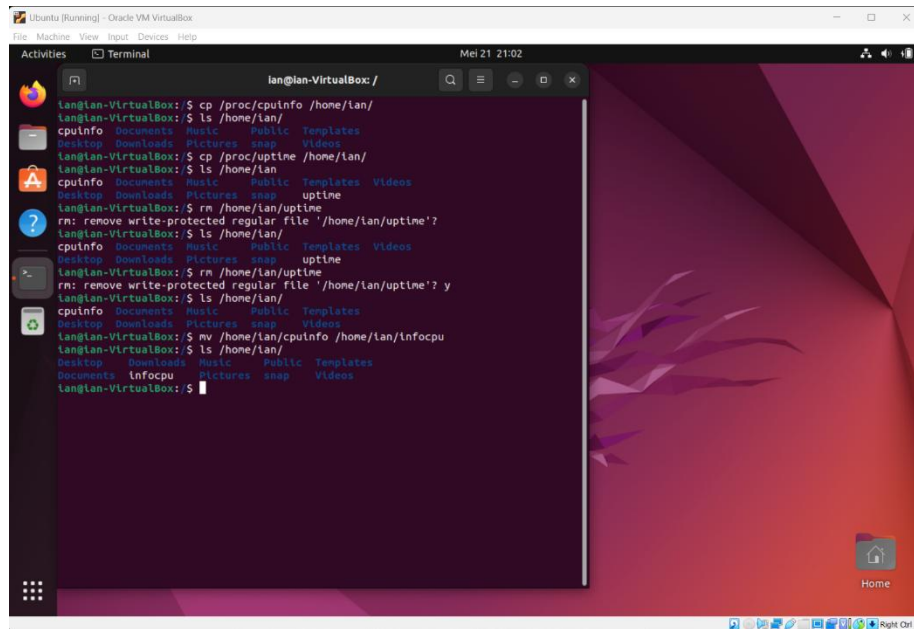
- Copy file /proc/uptime
cp /proc/uptime /home/ian
- Screenshot



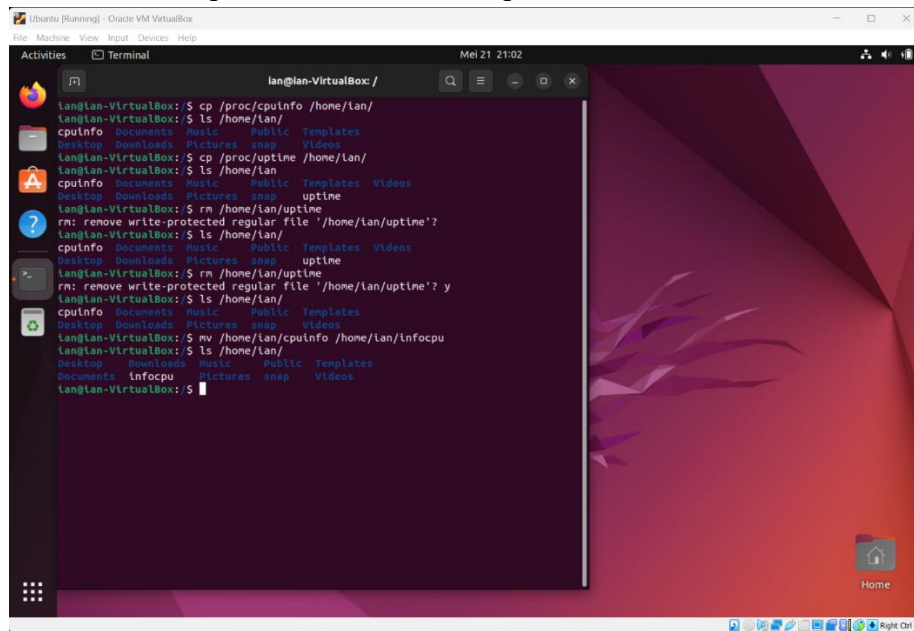
```
ian@ian-VirtualBox: /
langlan-VirtualBox: $ cp /proc/cpuinfo /home/ian/
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates
Desktop  Downloads  Pictures  snap  Videos
langlan-VirtualBox: $ cp /proc/uptime /home/ian/
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates  Videos
Desktop  Downloads  Pictures  snap  uptime
langlan-VirtualBox: $ rm /home/ian/uptime
rm: remove write-protected regular file '/home/ian/uptime'?
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates  Videos
Desktop  Downloads  Pictures  snap  uptime
langlan-VirtualBox: $ rm /home/ian/uptime
rm: remove write-protected regular file '/home/ian/uptime'? y
langlan-VirtualBox: $ ls /home/ian/
cpuinfo  Documents  Music  Public  Templates
Desktop  Downloads  Pictures  snap  Videos
langlan-VirtualBox: $ mv /home/ian/cpuinfo /home/ian/infocpu
langlan-VirtualBox: $ ls /home/ian/
Desktop  Downloads  Music  Public  Templates
Documents  infocpu  Pictures  snap  Videos
langlan-VirtualBox: $
```

- Hapus file uptime
rm /home/ian/uptime

f. Screenshot



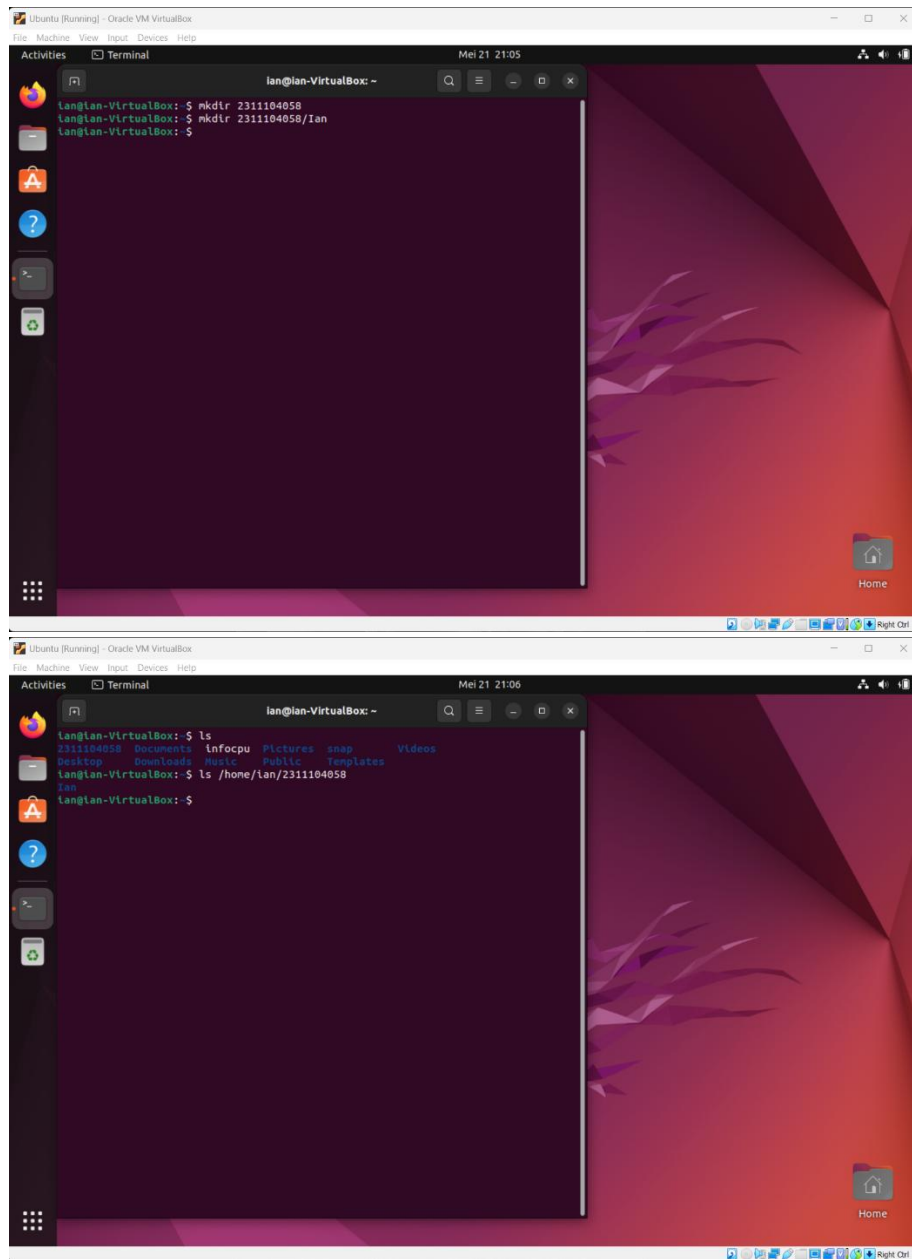
g. Rename file cpuinfo
mv /home/ian/cpuinfo /home/ian/infocpu



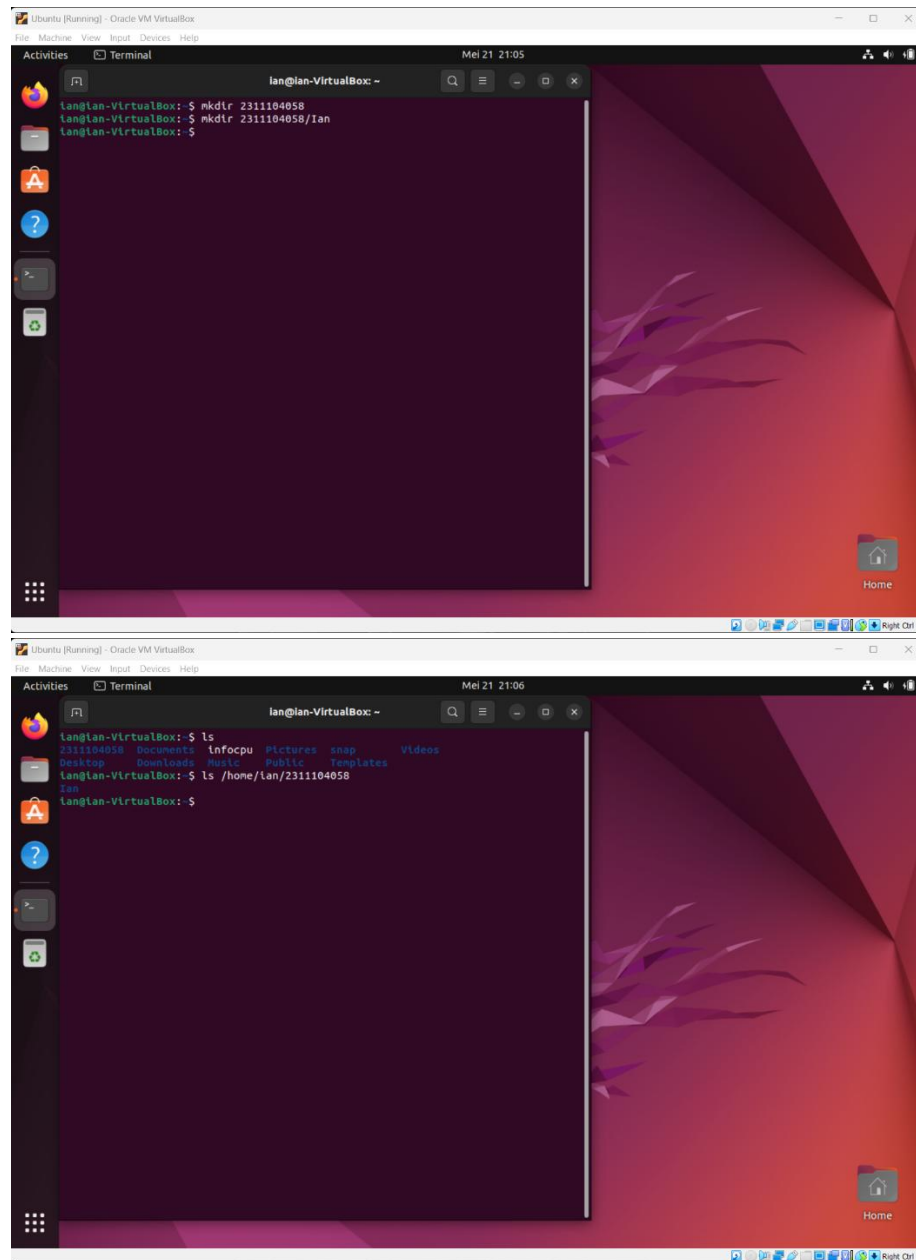
7. Membuat folder baru

a. Folder baru NIM

mkdir 2311104058

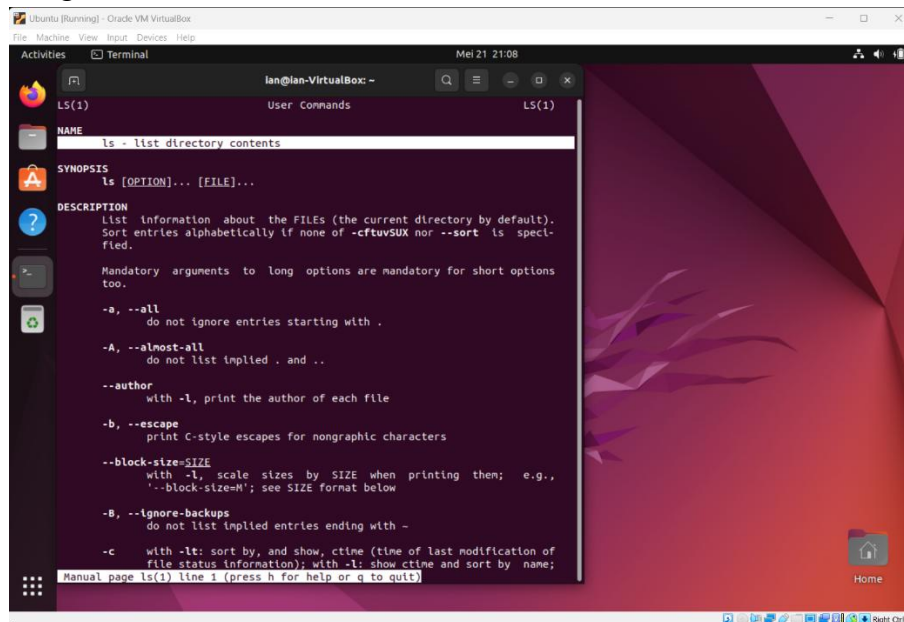


- b. Folder baru nama
mkdir ian



8. Membaca manual

a. Fungsi manual ls



```
lan@lan-VirtualBox: ~
ls - list directory contents
ls [OPTION]... [FILE]...

DESCRIPTION
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
fied.

Mandatory arguments to long options are mandatory for short options
too.

-a, --all
do not ignore entries starting with .

-A, --almost-all
do not list implied . and ..

--author
with -l, print the author of each file

-b, --escape
print C-style escapes for nongraphic characters

--block-size=SIZE
with -l, scale sizes by SIZE when printing them; e.g.,
'--block-size=M'; see SIZE format below

-B, --ignore-backups
do not list implied entries ending with ~

-c
with -lt: sort by, and show, ctime (time of last modification of
file status information); with -l: show ctime and sort by name;

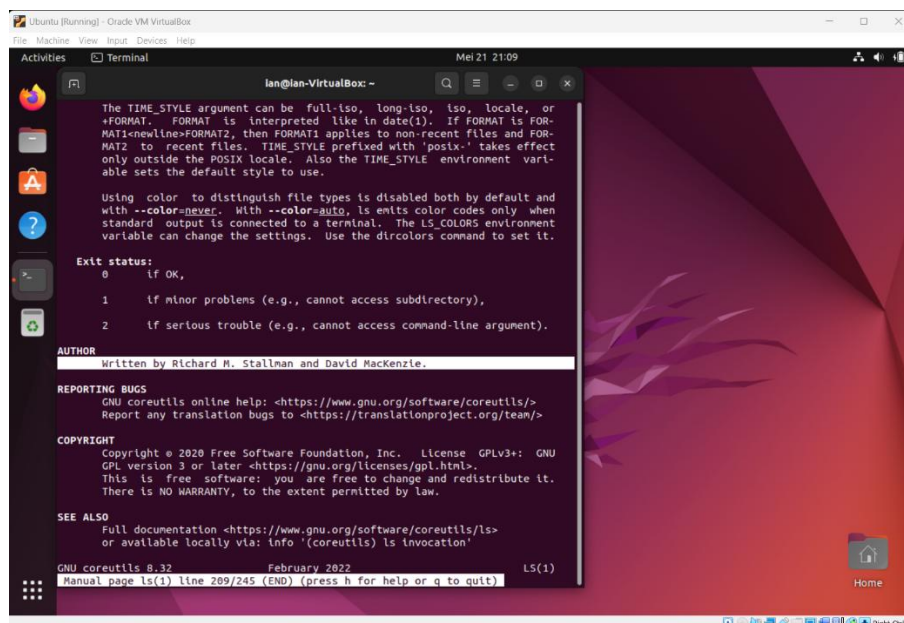
Manual page ls(1) line 1 (press h for help or q to quit)
```

b. Fungsi perintah ls

List directory contents untuk menampilkan daftar isi sebuah direktori

c. Pencipta perintah ls

Richard Stallman dan David MacKenzie



```
lan@lan-VirtualBox: ~
The TIME_STYLE argument can be full-iso, long-iso, iso, locale, or
+FORMAT. FORMAT is interpreted like in date(1). If FORMAT is FOR-
MAT=none+FORMAT2, then FORMAT1 applies to non-recent files and FOR-
MAT2 to recent files. TIME_STYLE prefixed with 'posix-' takes effect
only outside the POSIX locale. Also the TIME_STYLE environment vari-
able sets the default style to use.

Using color to distinguish file types is disabled both by default and
with --color=never. With --color=auto, ls emits color codes only when
standard output is connected to a terminal. The LS_COLORS environment
variable can change the settings. Use the dircolors command to set it.

Exit status:
0      if OK,
1      if minor problems (e.g., cannot access subdirectory),
2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR
Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Report any translation bugs to <https://translationproject.org/team/>

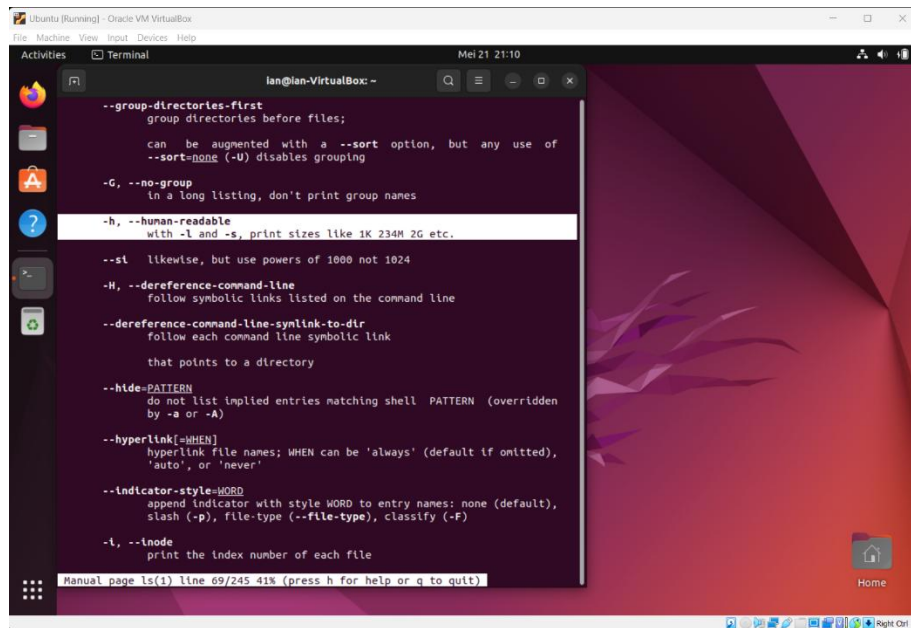
COPYRIGHT
Copyright © 2020 Free Software Foundation, Inc. License GPLv3+: GNU
GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

SEE ALSO
Full documentation <https://www.gnu.org/software/coreutils/ls>
or available locally via: info '(coreutils) ls invocation'

GNU coreutils 8.32 February 2022
Manual page ls(1) line 209/245 (END) (press h for help or q to quit)
```

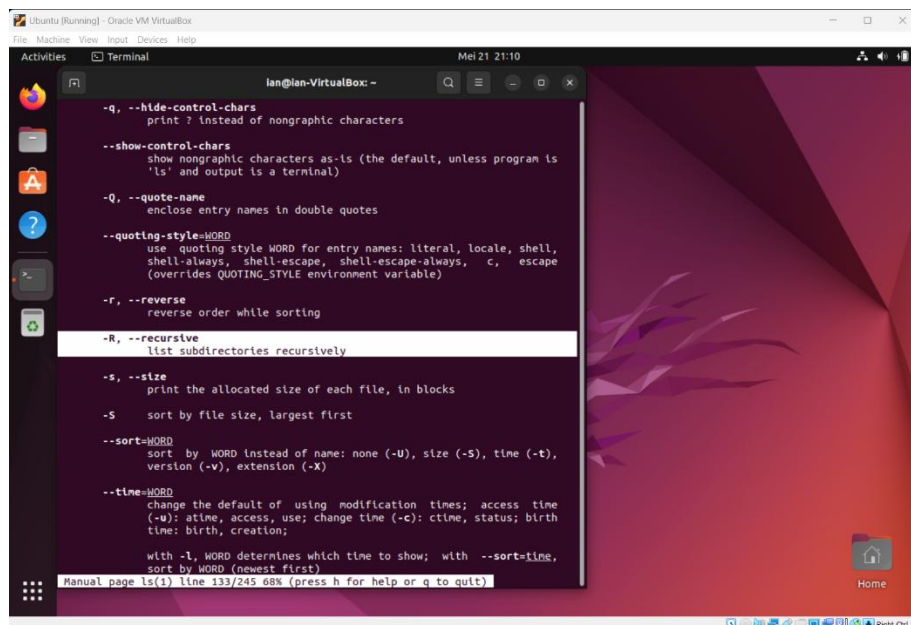
d. Arti dari -h dari manual ls

-h --human-readable jika digunakan bersama -l atau -s, ukuran file akan dicetak dalam format yang mudah dibaca manusia



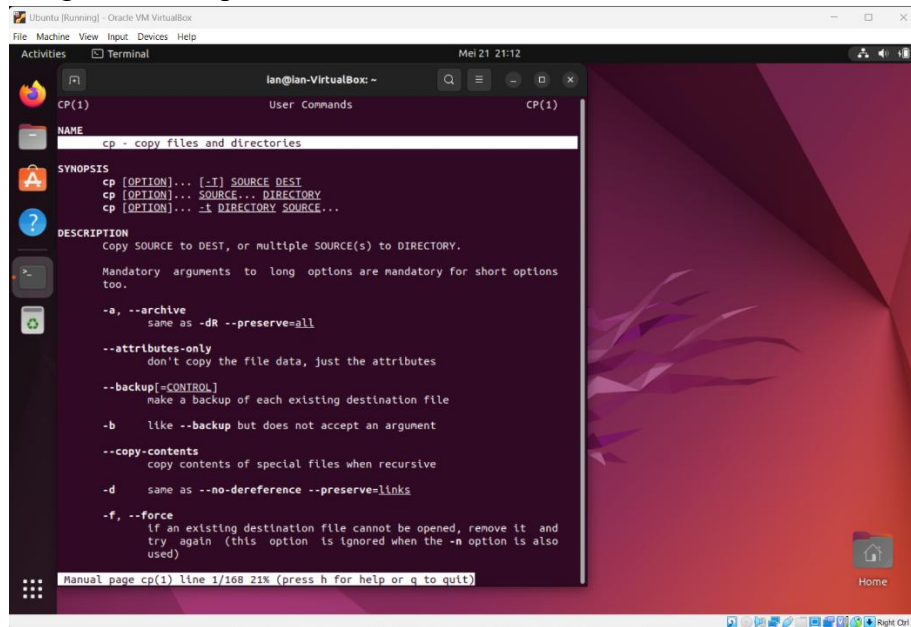
```
ian@ian-VirtualBox: ~  
--group-directories-first  
    group directories before files;  
    can be augmented with a --sort option, but any use of  
    --sort=none (-U) disables grouping  
-G, --no-group  
    in a long listing, don't print group names  
-h, --human-readable  
    with -l and -s, print sizes like 1K 234M 2G etc.  
--si  
    likewise, but use powers of 1000 not 1024  
-H, --dereference-command-line  
    follow symbolic links listed on the command line  
--dereference-command-line-symlink-to-dir  
    follow each command line symbolic link  
    that points to a directory  
--hide-PATTERN  
    do not list implied entries matching shell PATTERN (overridden  
    by -a or -A)  
--hyperlink[=WHEN]  
    hyperlink file names; WHEN can be 'always' (default if omitted),  
    'auto', or 'never'  
--indicator-style=WORD  
    append indicator with style WORD to entry names: none (default),  
    slash (-p), file-type (--file-type), classify (-F)  
-l, --inode  
    print the index number of each file  
Manual page ls(1) line 69/245 41% (press h for help or q to quit)
```

- e. Option rekursif
-R atau --recursive



```
ian@ian-VirtualBox: ~  
-Q, --hide-control-chars  
    print ? instead of nongraphic characters  
--show-control-chars  
    show nongraphic characters as-is (the default, unless program is  
    'ls' and output is a terminal)  
-Q, --quote-name  
    enclose entry names in double quotes  
--quoting-style=WORD  
    use quoting style WORD for entry names: literal, locale, shell,  
    shell-always, shell-escape, shell-escape-always, c, escape  
    (overrides QUOTING_STYLE environment variable)  
-r, --reverse  
    reverse order while sorting  
-R, --recursive  
    list subdirectories recursively  
-s, --size  
    print the allocated size of each file, in blocks  
-S  
    sort by file size, largest first  
--sort=WORD  
    sort by WORD instead of name: none (-U), size (-S), time (-t),  
    version (-V), extension (-X)  
--time=WORD  
    change the default of using modification times; access time  
    (-u): atime, access, use; change time (-c): ctime, status; birth  
    time: birth, creation;  
    with -l, WORD determines which time to show; with --sort=time,  
    sort by WORD (newest first)  
Manual page ls(1) line 133/245 68% (press h for help or q to quit)
```

f. Fungsi manual cp



The screenshot shows a terminal window titled "ian@ian-VirtualBox: ~" with a dark purple background. The terminal displays the manual page for the 'cp' command. The page is divided into sections: NAME, SYNOPSIS, DESCRIPTION, and a list of options. The NAME section states "cp - copy files and directories". The SYNOPSIS section shows the command syntax: "cp [OPTION]... [-I] SOURCE DEST", "cp [OPTION]... SOURCE... DIRECTORY", and "cp [OPTION]... -t DIRECTORY SOURCE...". The DESCRIPTION section explains that 'cp' copies SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY. It lists several options: -a, --archive (same as -dr --preserve=all), --attributes-only (don't copy the file data, just the attributes), --backup[=CONTROL] (make a backup of each existing destination file), -b (like --backup but does not accept an argument), --copy-contents (copy contents of special files when recursive), -d (same as --no-dereference --preserve=links), and -f, --force (if an existing destination file cannot be opened, remove it and try again (this option is ignored when the -n option is also used)). The bottom of the terminal shows "Manual page cp(1) line 1/168 21% (press h for help or q to quit)".

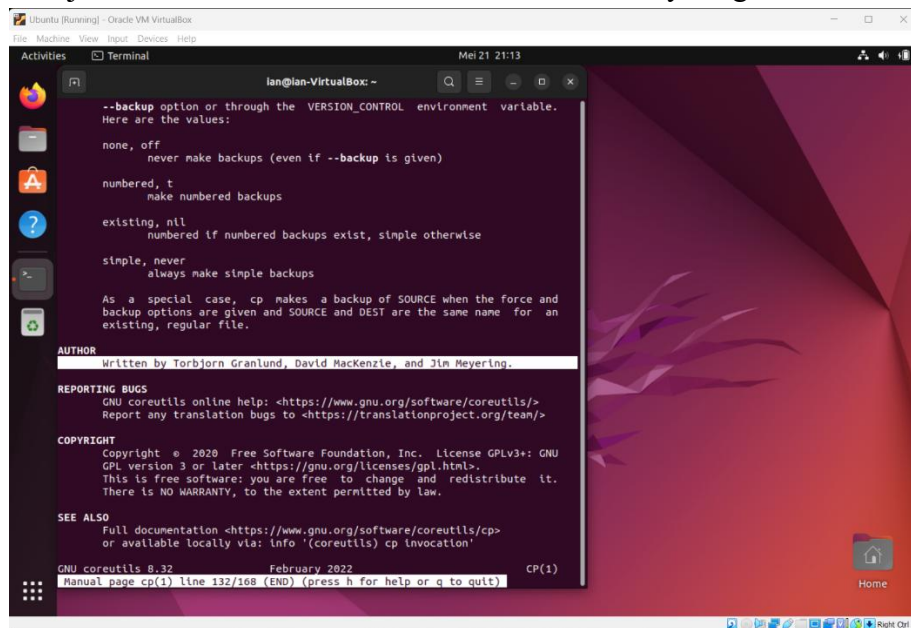
```
ian@ian-VirtualBox: ~  
CP(1)                                User Commands                                CP(1)  
NAME  
    cp - copy files and directories  
SYNOPSIS  
    cp [OPTION]... [-I] SOURCE DEST  
    cp [OPTION]... SOURCE... DIRECTORY  
    cp [OPTION]... -t DIRECTORY SOURCE...  
DESCRIPTION  
    Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.  
    Mandatory arguments to long options are mandatory for short options too.  
    -a, --archive  
        same as -dr --preserve=all  
    --attributes-only  
        don't copy the file data, just the attributes  
    --backup[=CONTROL]  
        make a backup of each existing destination file  
    -b  
        like --backup but does not accept an argument  
    --copy-contents  
        copy contents of special files when recursive  
    -d  
        same as --no-dereference --preserve=links  
    -f, --force  
        if an existing destination file cannot be opened, remove it and  
        try again (this option is ignored when the -n option is also  
        used)  
Manual page cp(1) line 1/168 21% (press h for help or q to quit)
```

g. Fungsi perintah cp

cp Copy files and directories (menyalin file dan direktori)

h. Pencipta perintah cp

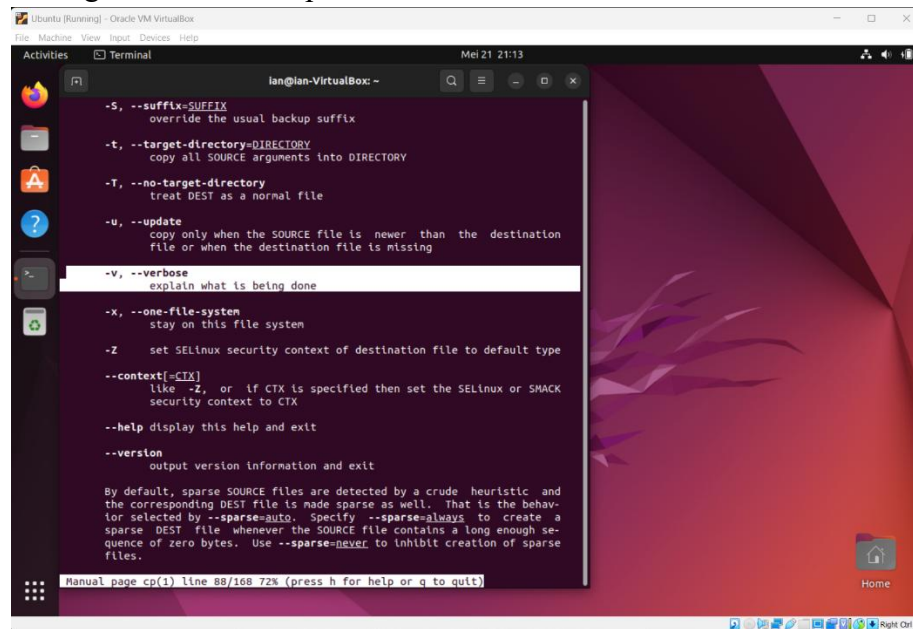
Torbjorn Granlund, David MacKenzie, dan Jim Meyering



The screenshot shows the continuation of the manual page for the 'cp' command in the same terminal window. It covers the backup options, author information, reporting bugs, copyright, and see also sections. The backup options section lists: none, off (never make backups (even if --backup is given)), numbered, t (make numbered backups), existing, nll (numbered if numbered backups exist, simple otherwise), and simple, never (always make simple backups). It also notes that 'cp' makes a backup of SOURCE when the force and backup options are given and SOURCE and DEST are the same name for an existing, regular file. The AUTHOR section states "Written by Torbjorn Granlund, David MacKenzie, and Jim Meyering." The REPORTING BUGS section provides links for GNU coreutils online help and translation bugs. The COPYRIGHT section states "Copyright © 2020 Free Software Foundation, Inc. License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>." and "This is free software: you are free to change and redistribute it. There is NO WARRANTY, to the extent permitted by law." The SEE ALSO section points to full documentation. The bottom of the terminal shows "GNU coreutils 8.32 February 2022" and "Manual page cp(1) line 132/168 (END) (press h for help or q to quit)".

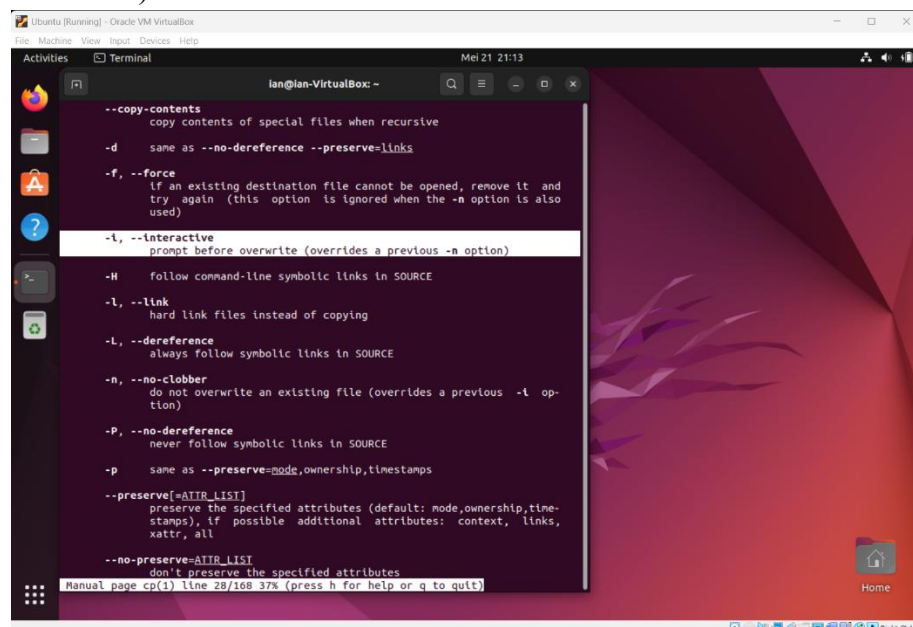
```
ian@ian-VirtualBox: ~  
    --backup option or through the VERSION_CONTROL environment variable.  
    Here are the values:  
    none, off  
        never make backups (even if --backup is given)  
    numbered, t  
        make numbered backups  
    existing, nll  
        numbered if numbered backups exist, simple otherwise  
    simple, never  
        always make simple backups  
    As a special case, cp makes a backup of SOURCE when the force and  
    backup options are given and SOURCE and DEST are the same name for an  
    existing, regular file.  
AUTHOR  
    Written by Torbjorn Granlund, David MacKenzie, and Jim Meyering.  
REPORTING BUGS  
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>  
    Report any translation bugs to <https://translationproject.org/team/>  
COPYRIGHT  
    Copyright © 2020 Free Software Foundation, Inc. License GPLv3+: GNU  
    GPL version 3 or later <https://gnu.org/licenses/gpl.html>.  
    This is free software: you are free to change and redistribute it.  
    There is NO WARRANTY, to the extent permitted by law.  
SEE ALSO  
    Full documentation <https://www.gnu.org/software/coreutils/cp>  
    or available locally via: info '(coreutils) cp invocation'  
GNU coreutils 8.32 February 2022  
Manual page cp(1) line 132/168 (END) (press h for help or q to quit)
```


- i. Arti -v dalam perintah cp
--verbose, berfungsi untuk menjelaskan atau menampilkan status dari apa yang sedang dilakukan oleh perintah salin tersebut



```
lan@lan-VirtualBox: ~  
-S, --suffix=SUFFIX  
    override the usual backup suffix  
-t, --target-directory=DIRECTORY  
    copy all SOURCE arguments into DIRECTORY  
-T, --no-target-directory  
    treat DEST as a normal file  
-u, --update  
    copy only when the SOURCE file is newer than the destination  
    file or when the destination file is missing  
-v, --verbose  
    explain what is being done  
-x, --one-file-system  
    stay on this file system  
-Z  
    set SELinux security context of destination file to default type  
--context=CTX  
    like -Z, or if CTX is specified then set the SELinux or SMACK  
    security context to CTX  
--help  
    display this help and exit  
--version  
    output version information and exit  
  
By default, sparse SOURCE files are detected by a crude heuristic and  
the corresponding DEST file is made sparse as well. That is the behav-  
ior selected by --sparse=auto. Specify --sparse=always to create a  
sparse DEST file whenever the SOURCE file contains a long enough se-  
quence of zero bytes. Use --sparse=never to inhibit creation of sparse  
files.  
  
Manual page cp(1) line 88/168 72% (press h for help or q to quit)
```

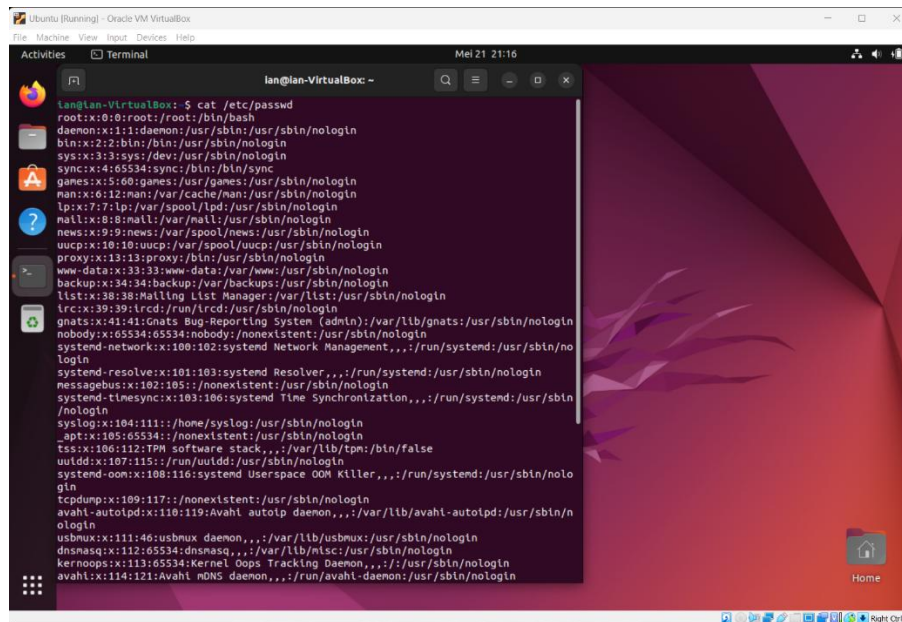
- j. Option interaktif
-i atau --interactive (meminta konfirmasi/prompt sebelum menimpa file yang sudah ada)



```
lan@lan-VirtualBox: ~  
--copy-contents  
    copy contents of special files when recursive  
-d  
    same as --no-dereference --preserve=links  
-f, --force  
    if an existing destination file cannot be opened, remove it and  
    try again (this option is ignored when the -n option is also  
    used)  
-i, --interactive  
    prompt before overwrite (overrides a previous -n option)  
-H  
    follow command-line symbolic links in SOURCE  
-l, --link  
    hard link files instead of copying  
-L, --dereference  
    always follow symbolic links in SOURCE  
-n, --no-clobber  
    do not overwrite an existing file (overrides a previous -i op-  
    tion)  
-P, --no-dereference  
    never follow symbolic links in SOURCE  
-p  
    same as --preserve=mode,ownership,timestamps  
--preserve=ATTR_LIST  
    preserve the specified attributes (default: mode,ownership,time-  
    stamps), if possible additional attributes: context, links,  
    xattr, all  
--no-preserve=ATTR_LIST  
    don't preserve the specified attributes  
  
Manual page cp(1) line 28/168 37% (press h for help or q to quit)
```

9. Pipe

a. Screenshot

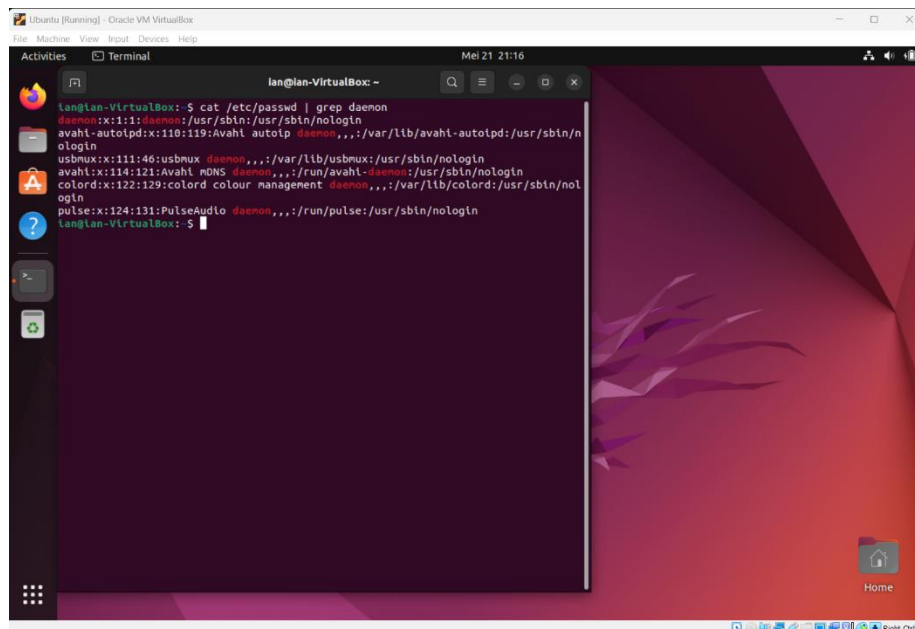


```
lan@lan-VirtualBox: ~  
$ cat /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mail Manager:/var/llist:/usr/sbin/nologin  
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin  
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:100:102:systemd Network Management,,:/run/systemd:/usr/sbin/nologin  
systemd-resolve:x:101:103:systemd Resolver,,:/run/systemd:/usr/sbin/nologin  
messagebus:x:182:185:,:/nonexistent:/usr/sbin/nologin  
systemd-timesync:x:103:106:systemd Time Synchronization,,:/run/systemd:/usr/sbin/nologin  
syslog:x:104:111:,:/home/syslog:/usr/sbin/nologin  
_apt:x:105:65534:,:/nonexistent:/usr/sbin/nologin  
tss:x:106:112:TPM software stack,,:/var/lib/tpm:/bin/false  
uuidd:x:107:115:,:/run/uuidd:/usr/sbin/nologin  
systemd-oom:x:108:116:systemd Userspace OOM Killer,,:/run/systemd:/usr/sbin/nologin  
tcpdump:x:109:117:,:/nonexistent:/usr/sbin/nologin  
avahi-autoipd:x:110:119:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin  
usbmux:x:111:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin  
dnsmasq:x:112:65534:dnsmasq,,:/var/lib/misc:/usr/sbin/nologin  
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,:/usr/sbin/nologin  
avahi:x:114:121:Avahi mDNS daemon,,:/run/avahi-daemon:/usr/sbin/nologin
```

b. Fungsi perintah cat

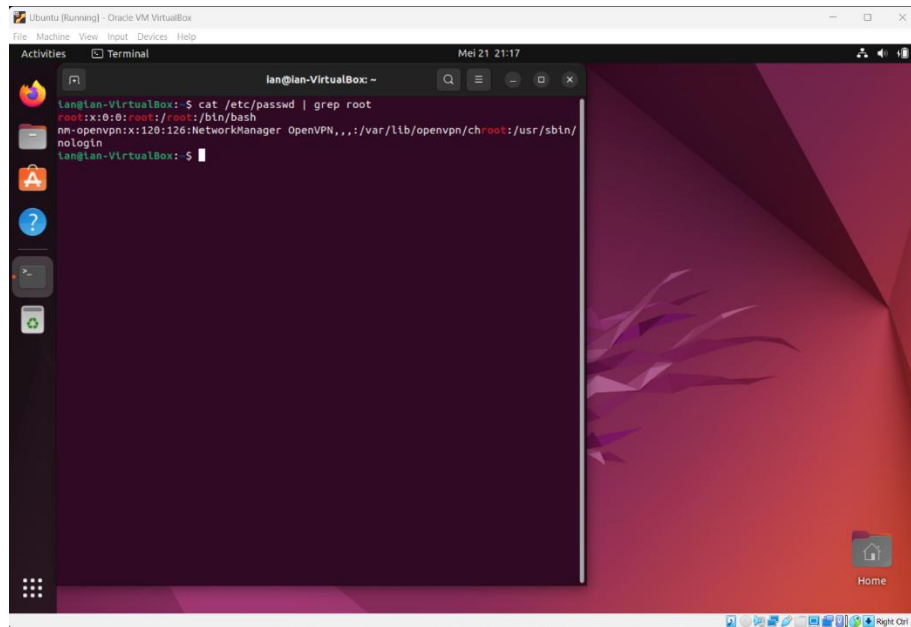
Menyatukan (concatenate) isi file dan menampilkannya sebagai standard output ke layar terminal.

c. Screenshot



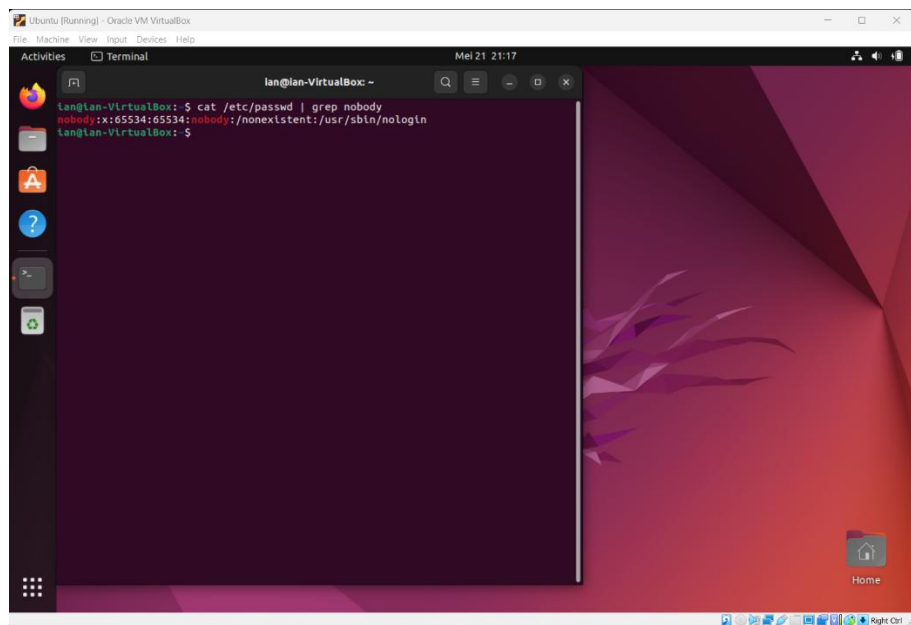
```
lan@lan-VirtualBox: ~  
$ cat /etc/passwd | grep daemon  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
avahi-autoipd:x:110:119:Avahi autoip daemon,,:/var/lib/avahi-autoipd:/usr/sbin/nologin  
usbmux:x:111:46:usbmux daemon,,:/var/lib/usbmux:/usr/sbin/nologin  
avahi:x:114:121:Avahi mDNS daemon,,:/run/avahi-daemon:/usr/sbin/nologin  
colord:x:122:129:colord colour management daemon,,:/var/lib/colord:/usr/sbin/nologin  
pulse:x:124:131:PulseAudio daemon,,:/run/pulse:/usr/sbin/nologin  
lan@lan-VirtualBox: ~
```

d. Screenshot



```
lan@lan-VirtualBox: ~$ cat /etc/passwd | grep root
root:x:0:0:root:/root:/bin/bash
nn-openvpn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
lan@lan-VirtualBox: ~$
```

e. Screenshot



```
lan@lan-VirtualBox: ~$ cat /etc/passwd | grep nobody
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
lan@lan-VirtualBox: ~$
```

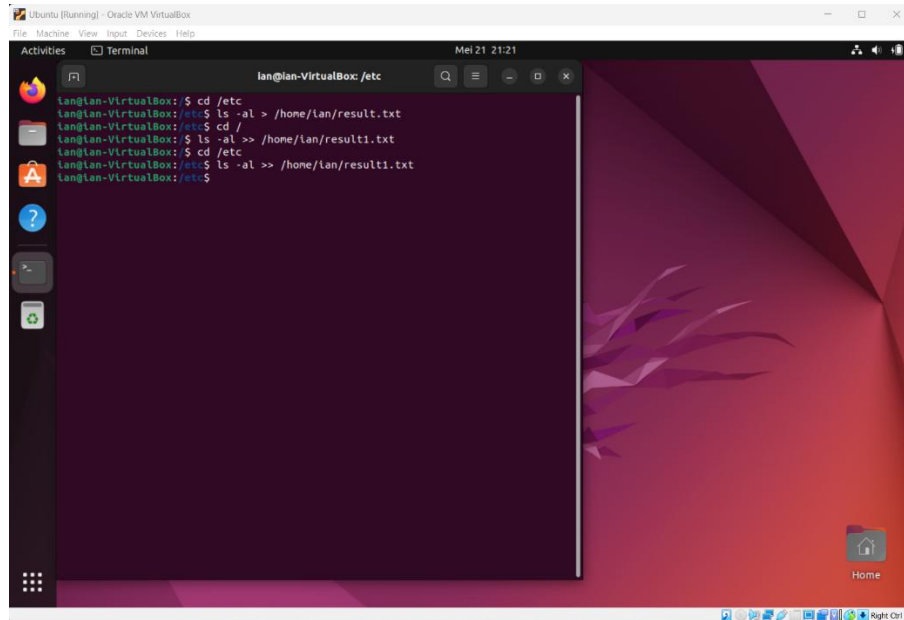
f. Fungsi perintah “| grep daemon”

Simbol | (pipe) berfungsi untuk mengambil output dari perintah sebelumnya (dalam hal ini cat) dan menjadikannya sebagai input untuk perintah selanjutnya. Perintah grep daemon berfungsi untuk menyaring (filter) input tersebut dan hanya menampilkan baris teks yang mengandung kata "daemon".

10. Redirection

a. Screenshot dan penjelasan

Penjelasan `ls -al > /home/ian/result.txt`: Hasil daftar direktori / tidak akan muncul di layar. Tanda `>` akan mengarahkan dan menuliskan hasil (output) perintah tersebut ke dalam sebuah file bernama `result.txt`.



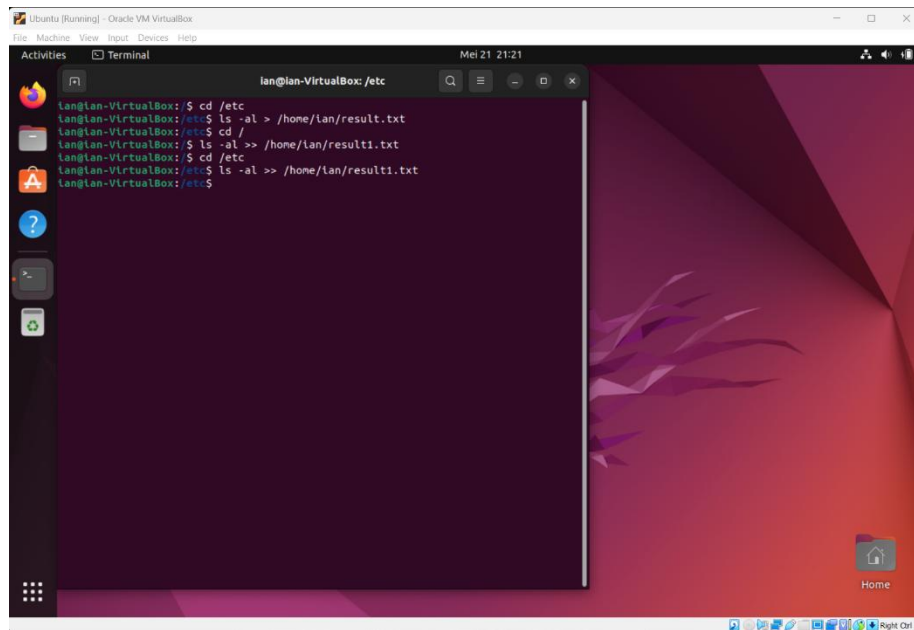
```
ian@ian-VirtualBox: /etc
ianlan-VirtualBox:/$ cd /etc
ianlan-VirtualBox:/etc$ ls -al > /home/ian/result.txt
ianlan-VirtualBox:/etc$ cd /
ianlan-VirtualBox:/$ ls -al >> /home/ian/result1.txt
ianlan-VirtualBox:/$ cd /etc
ianlan-VirtualBox:/etc$ ls -al >> /home/ian/result1.txt
ianlan-VirtualBox:/etc$
```

b. Letak file `result.txt`

Di dalam direktori `/home/ian/`

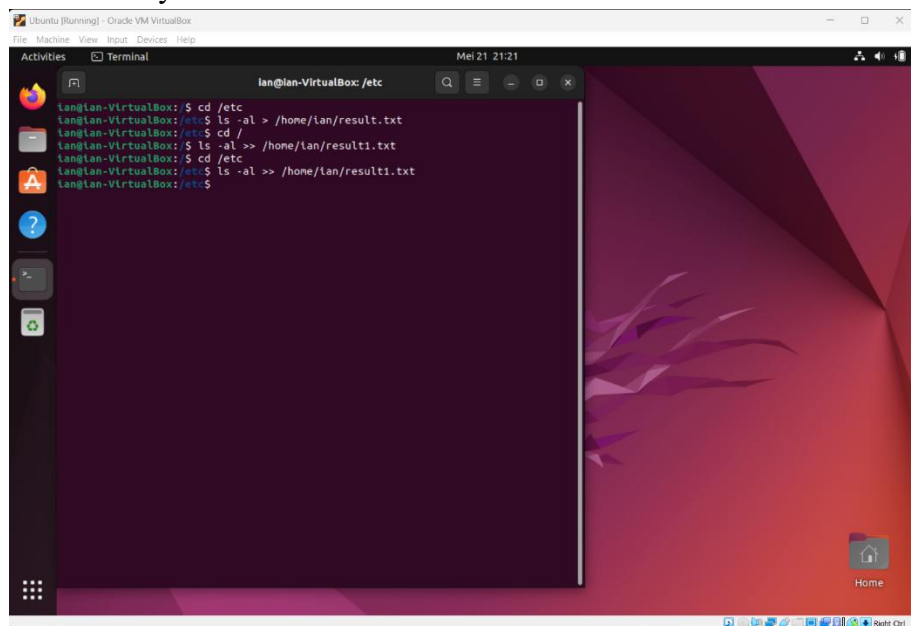
c. Screenshot dan penjelasan

Penjelasan `ls -al > /home/ian/result.txt` di `/etc`: Tanda `>` akan menimpa (overwrite) isi file `result.txt` yang lama dengan hasil daftar direktori dari `/etc`. Isi awal tentang direktori / akan hilang.



```
lan@lan-VirtualBox: /etc
lan@lan-VirtualBox:~$ cd /etc
lan@lan-VirtualBox:/etc$ ls -al > /home/lan/result.txt
lan@lan-VirtualBox:/etc$ cd /
lan@lan-VirtualBox:/$$ ls -al >> /home/lan/result1.txt
lan@lan-VirtualBox:/$$ cd /etc
lan@lan-VirtualBox:/etc$ ls -al >> /home/lan/result1.txt
lan@lan-VirtualBox:/etc$
```

- d. Fungsi perintah >
- Merupakan operator redirection yang mengalihkan standard output ke sebuah file. Jika file belum ada, ia akan membuatnya; jika file sudah ada, ia akan menimpa seluruh isinya.

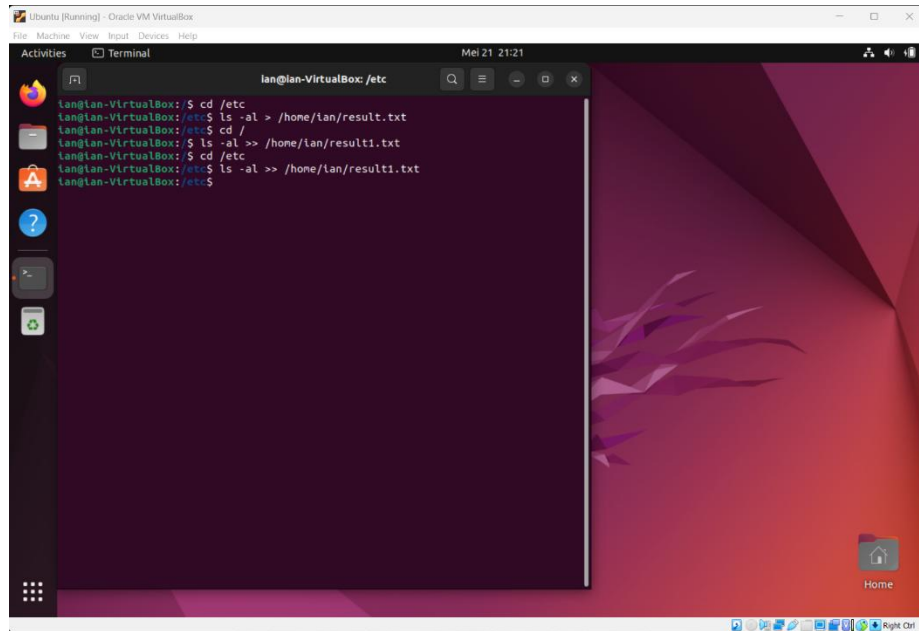


```
lan@lan-VirtualBox: /etc
lan@lan-VirtualBox:~$ cd /etc
lan@lan-VirtualBox:/etc$ ls -al > /home/lan/result.txt
lan@lan-VirtualBox:/etc$ cd /
lan@lan-VirtualBox:/$$ ls -al >> /home/lan/result1.txt
lan@lan-VirtualBox:/$$ cd /etc
lan@lan-VirtualBox:/etc$ ls -al >> /home/lan/result1.txt
lan@lan-VirtualBox:/etc$
```

- e. Screenshot dan penjelasan
- Sama seperti poin a, hasil disimpan di result1.txt

f. Screenshot dan penjelasan

Karena menggunakan >>, hasil dari folder /etc akan ditambahkan (append) di baris paling bawah dari file result1.txt tanpa menghapus isi sebelumnya.

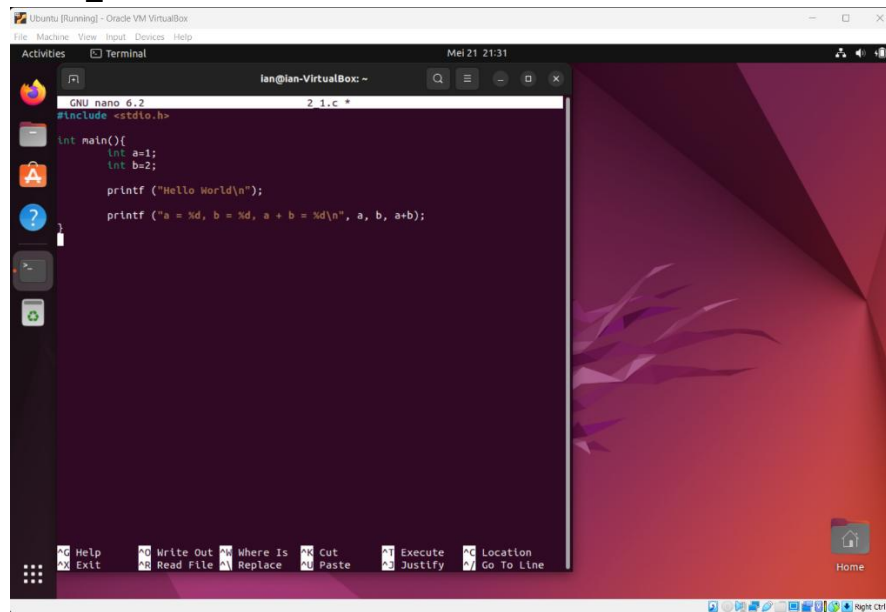


g. Perbedaan > dan >>

> berfungsi untuk membuat atau menimpa (overwrite) file, sedangkan >> berfungsi untuk menambahkan (append) konten baru ke bagian akhir file yang sudah ada tanpa menghapus isinya

II. Compile Source Code

1. File 2_1.c

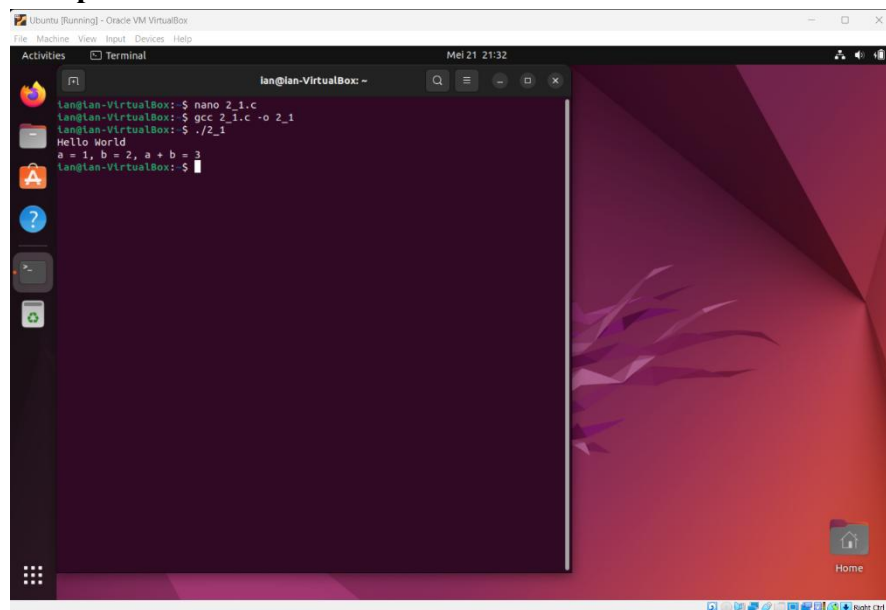


```
GNU nano 6.2 2_1.c
#include <stdio.h>

int main(){
    int a=1;
    int b=2;

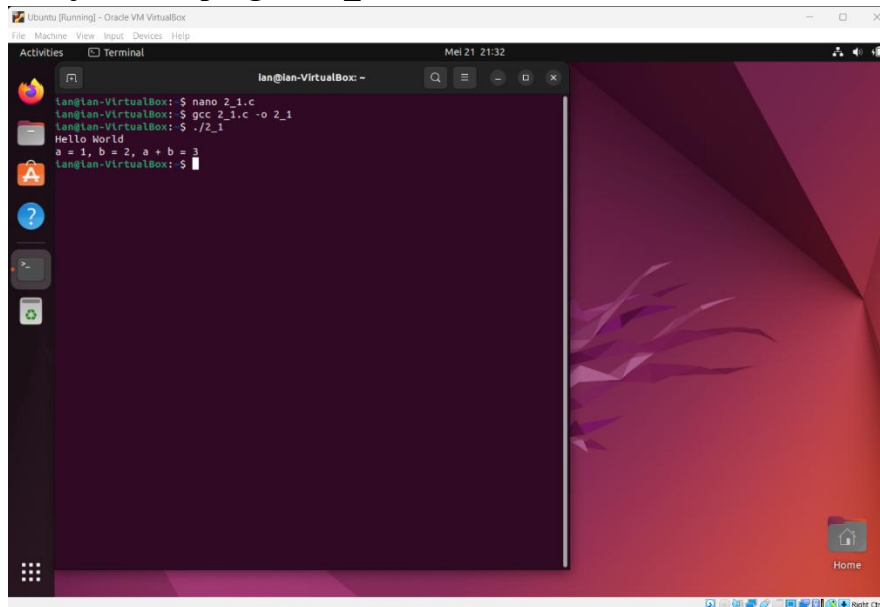
    printf ("Hello World\n");
    printf ("a = %d, b = %d, a + b = %d\n", a, b, a+b);
}
```

2. Compile



```
langlan-VirtualBox: ~
langlan-VirtualBox: $ nano 2_1.c
langlan-VirtualBox: $ gcc 2_1.c -o 2_1
langlan-VirtualBox: $ ./2_1
Hello World
a = 1, b = 2, a + b = 3
langlan-VirtualBox: $
```

3. Menjalankan program 2_1.c

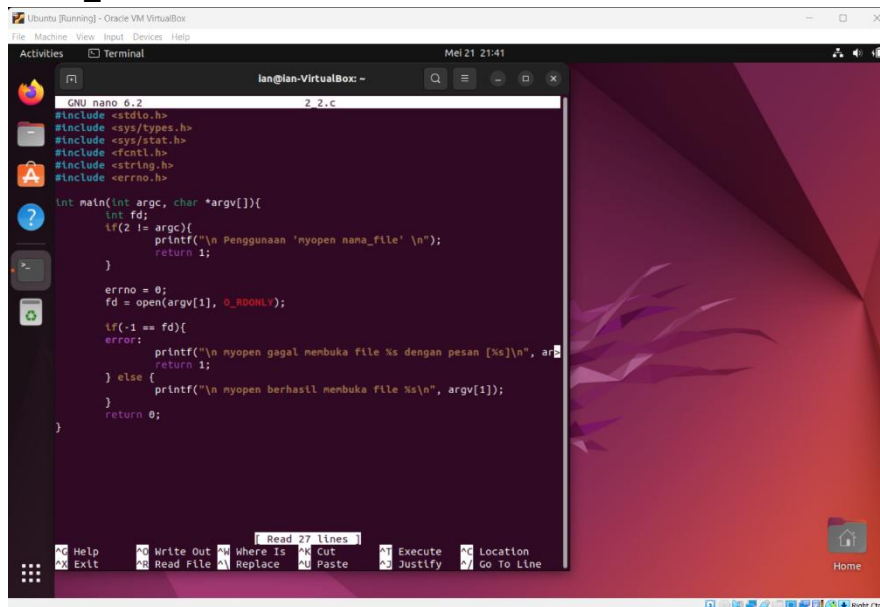


The screenshot shows a terminal window titled 'lan@lan-VirtualBox: ~' with the following commands and output:

```
lan@lan-VirtualBox: ~$ nano 2_1.c
lan@lan-VirtualBox: ~$ gcc 2_1.c -o 2_1
lan@lan-VirtualBox: ~$ ./2_1
Hello World
a = 1, b = 2, a + b = 3
lan@lan-VirtualBox: ~$
```

The terminal output shows the program successfully compiled and executed, printing 'Hello World' and the sum of a and b.

4. File 2_2.c



The screenshot shows a terminal window titled 'lan@lan-VirtualBox: ~' with the following code in a file named 2_2.c:

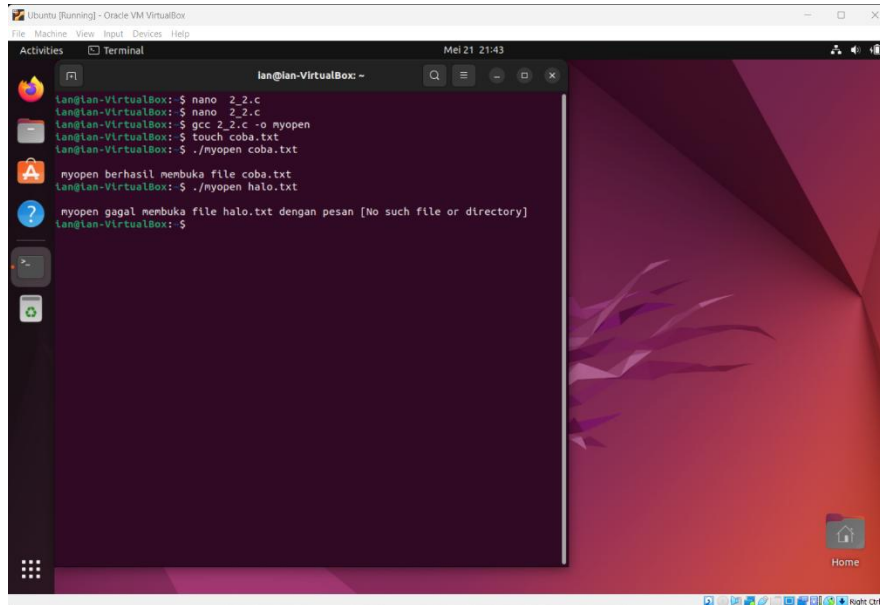
```
GNU nano 6.2 2_2.c
#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <string.h>
#include <errno.h>

int main(int argc, char *argv[]){
    int fd;
    if(2 != argc){
        printf("\n Penggunaan 'nyopen nana_file' \n");
        return 1;
    }

    errno = 0;
    fd = open(argv[1], O_RDONLY);
    if(-1 == fd){
        error:
        printf("\n nyopen gagal membuka file %s dengan pesan [%s]\n", argv[1], strerror(errno));
        return 1;
    } else {
        printf("\n nyopen berhasil membuka file %s\n", argv[1]);
    }
    return 0;
}
```

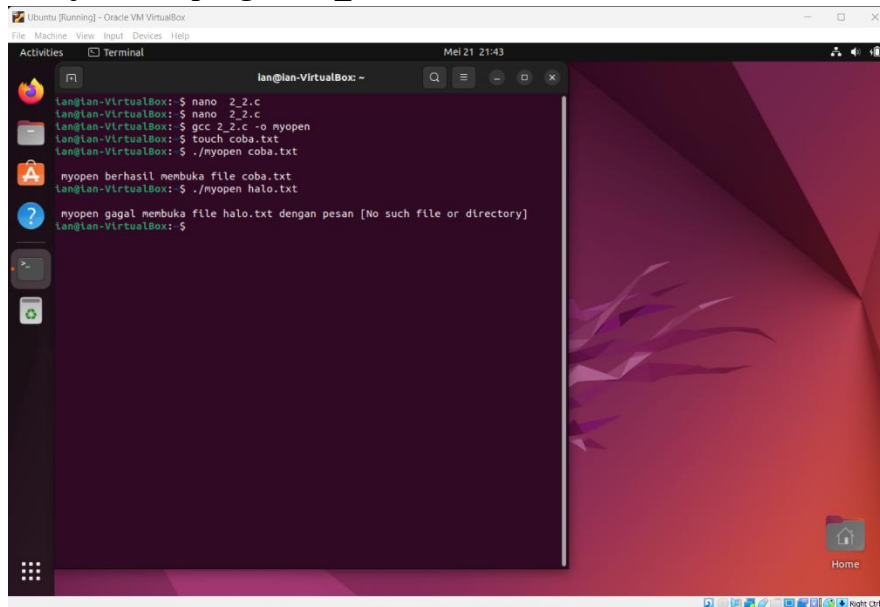
The code is a C program that takes a filename as an argument and attempts to open it in read-only mode. It includes error handling for file opening failures.

5. Compile



```
lan@lan-VirtualBox: ~  
lan@lan-VirtualBox: $ nano 2_2.c  
lan@lan-VirtualBox: $ nano 2_2.c  
lan@lan-VirtualBox: $ gcc 2_2.c -o nyopen  
lan@lan-VirtualBox: $ touch coba.txt  
lan@lan-VirtualBox: $ ./nyopen coba.txt  
nyopen berhasil membuka file coba.txt  
lan@lan-VirtualBox: $ ./nyopen halo.txt  
nyopen gagal membuka file halo.txt dengan pesan [No such file or directory]  
lan@lan-VirtualBox: $
```

6. Menjalankan program 2_2.c



```
lan@lan-VirtualBox: ~  
lan@lan-VirtualBox: $ nano 2_2.c  
lan@lan-VirtualBox: $ nano 2_2.c  
lan@lan-VirtualBox: $ gcc 2_2.c -o nyopen  
lan@lan-VirtualBox: $ touch coba.txt  
lan@lan-VirtualBox: $ ./nyopen coba.txt  
nyopen berhasil membuka file coba.txt  
lan@lan-VirtualBox: $ ./nyopen halo.txt  
nyopen gagal membuka file halo.txt dengan pesan [No such file or directory]  
lan@lan-VirtualBox: $
```

7. Penjelasan

Program ini digunakan untuk mensimulasikan proses pembukaan sebuah file menggunakan system call. Pertama, program akan mengecek jumlah argumen; jika tidak ada nama file yang dimasukkan, program akan menampilkan panduan penggunaan dan berhenti. Jika nama file diberikan, program mencoba membuka file tersebut dengan mode Read-Only (O_RDONLY). Jika file gagal dibuka (misalnya file tidak ditemukan), program akan mencetak pesan error sesuai dengan kesalahan

sistemnya. Sebaliknya, jika berhasil, ia akan mencetak pesan bahwa program "berhasil membuka file"